XINRAN (SARAH) HU

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Purdue University
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EDUCATION

Purdue University, West Lafayette, Indiana, US — Expected graduation in May 2027
Doctor of Philosophy, Construction Management Technology
Advisor: Dr. Yunfeng (Cindy) Chen
Dissertation: A Computer Vision and Multi-Sensor Framework for Human Pose Estimation and Integrated Workplace Risk Assessment in Field Maintenance Tasks (In progress)

Wuhan University, Wuhan, Hubei, China — June 2022
Master of Science in Civil Engineering, Project Management
Advisor: Dr. Lin Yang
Thesis: Research on Dynamic Evolution Mechanism of Inter-organization Field in Complex Projects

Wuhan University, Wuhan, Hubei, China — June 2019
Bachelor of Science, Civil Engineering
Advisor: Dr. Lin Yang
Thesis: Organizational Synchronization Strategies for Addressing Construction Delays in Complex Projects

ACADEMIC EMPLOYMENT

Bowen School of Construction, Purdue University — Aug 2022 – Present
- Research Assistant
- Teaching Assistant

School of Civil Engineering, Wuhan University — Jan 2020 – Dec 2020
- Teaching Assistant

INDUSTRY INTERNSHIPS

China Construction Eighth Engineering Division (CSCEC), Fujian, China — Jul 2018 – Aug 2018
- Construction Engineering Intern, Nandaiyunshan Tunnel, Nanping–Longyan Railway
- Observed tunnel construction operations, site management, and safety practices

Duyi Project Management Co., Yancheng, Jiangsu, China — Jul 2017
- Project Management Intern

- Observed cost audits of building finishing works and visited active construction sites

Jianxiong Electric Group, Yancheng, Jiangsu, China Jul 2017
- Construction Management Intern
- Assisted with bid document preparation and observed contract and safety management practices

RESEARCH INTERESTS

Construction Safety | Human-Computer Interaction | Human Factors | Artificial Intelligence | Computer Vision | Wearable Sensing | Workforce Development | Work Zone Safety | Building Information Modeling

PUBLICATIONS

JOURNAL ARTICLES

Published

[J02] Yang, L., **Hu, X.,** & Zhao, X. (2025). Influence on BIM project performance at project life-cycle stages: A network perspective. *Journal of Building Design and Environment*, *3*(3), 202578-202578. https://doi.org/10.70401/jbde.2025.0022

[J01] Yang, L., **Hu, X.,** & Zhao, X. (2022). Organization synchronization in response to complex project delays: network-based analysis. *Buildings*, *12*(5), 662. https://doi.org/10.3390/buildings12050662

To Be Submitted & In Preparation

Hu, X., Guo, X., Chen, Y., & Zhang, J. (2026). Effects of an Add-on Ergonomic Handle on Trunk and Shoulder Kinematics During Simulated Highway Maintenance Shoveling: A Markerless 3D Motion Capture Study. *Applied Ergonomics*, to be submitted.

Hu, X., Guo, X., Chen, Y., & Zhang, J. (2026). Effects of Ergonomic Handle and Back Exoskeleton on Physical Fatigue during Shoveling: Physiological and Subjective Responses. *Applied Ergonomics*, to be submitted.

Hu, X., Guo, X., Chen, Y., & Zhang, J. (2026). Accuracy of Markerless Human Pose Estimation Models in Field Transportation Maintenance Work: A Validation Study Using Xsens Motion Capture. *Automation in Construction*, in preparation.

CONFERENCE PROCEEDINGS

[C11] Hu, X., Guo, X., Chen, Y., & Zhang, J. (2026). Impact of Body Mass Index, Tenure Length, and Age on Physical Exertion of Transportation Maintenance Workers during Simulated Asphalt Shoveling. *2026 AHFE International Conference on Human Factors in Design, Engineering, and Computing, Honolulu, Hawaii, US. [Full draft completed]*

[C10] Hu, X., Guo, X., Chen, Y., & Zhang, J. (2025). Usability of Passive Back Exoskeletons and Ergonomic Handles for Transportation Maintenance Workers: A Field-Based Study

in Shoveling. In *CIB Conferences* (Vol. 1, No. 1, p. 175). https://doi.org/10.7771/3067-4883.1930

[C09] Hu, X., Guo, X., Chen, Y., & Zhang, J. (2025). The impacts of back exoskeletons and ergonomic handles on joint mechanics during shoveling activities. *Transportation Research Board 2025,* Washington DC, US. [Abstract only]

[C08] Hu, X., Guo, X., Chen, Y., & Zhang, J. (2024). Video-based ergonomic risk assessment among transportation maintenance workers in shoveling. *2024 AHFE International Conference on Human Factors in Design, Engineering, and Computing*, Honolulu, Hawaii, US. http://doi.org/10.54941/ahfe1005718

[C07] Hu, X., Guo, X., Chen, Y., & Zhang, J. (2024, August). Classification of Physical Fatigue Levels in Shoveling Using Heart Rate, Electrodermal Data, and Skin Temperature. In *International Conference on Computing in Civil and Building Engineering* (pp. 259-269). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-84224-5_21

[C06] Guo, X., **Hu, X.,** Chen, Y., & Zhang, J. Assessing Physical Fatigue in Highway Maintenance Workers through Electromyography: The Impact of Lifting Weights and Heights on Bag Handling. In *Computing in Civil Engineering 2024* (pp. 821-829). https://doi.org/10.1061/9780784486139.088

[C05] Hu, X., Guo, X., & Chen, Y. (2024). BIM's Impact on Project Management Performance. *Construction Research Congress 2024* (pp. 711-721), Des Moines, Iowa, US. https://doi.org/10.1061/9780784485286.07

[C04] Guo, X., **Hu, X.,** & Chen, Y. (2023). Exploration of back exoskeleton's effectiveness on transportation maintenance workers during lifting activities. *14th International Conference on Applied Human Factors and Ergonomics 2023: Physical Ergonomics and Human Factors*, San Francisco, California, US. https://dx.doi.org/10.54941/ahfe1003044

[C03] Hu, X., & Chen, Y. (2023). Impact of BIM on project performance. *Proceedings of the 2023 European Conference on Computing in Construction and the 40th International CIB W78 Conference*, *4*(1–8). https://doi.org/10.35490/EC3.2023.293

[C02] Yang, L., Lou, J., & **Hu, X.** (2021, November). Increasing Resilience of Utility Tunnel PPP Projects Through Risk Management: A Case on in Shiyan City. In *International Symposium on Advancement of Construction Management and Real Estate* (pp. 1577-1591). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-19-5256-2_121

[C01] Yang, L., **Hu, X.,** & Lou, J. (2020, November). Organizational interaction and dilemma governance strategies in response to the COVID-19 epidemic. In *International Symposium on Advancement of Construction Management and Real Estate* (pp. 85-105). Singapore: Springer Singapore. https://doi.org/10.1007/978-981-16-3587-8_8

TECHNICAL REPORTS

[R01] Dunlop, S. R., Ukkusuri, S. V., Chen, Y., Thakkar, D. J., Ka, E., Mittal, S., **Hu, X.,** Wang, Z., Khanna, I., & Tammineni, R. (2026). *INDOT training gap analysis, developing a training program for INDOT workforce development (new, front-line, supervisors) and smart tool for effective planning for workforce*. (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2026/12). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318620

PRESENTATIONS & POSTERS

[P04] Hu, X., Guo, X., Chen, Y., & Zhang, J. (2025). The impacts of back exoskeletons and ergonomic handles on joint mechanics during shoveling activities. *Transportation Research Board 2025,* Washington DC, US. [Lecture presentation]

[P03] Guo, X., **Hu, X.,** Chen, Y., & Zhang, J. (2024). Applied Ergonomics. *Construction Advisory Council 2024,* West Lafayette, Indiana US.

[P02] Guo, X., **Hu, X.,** Chen, Y., & Zhang, J. (2023, 2024, 2025, & 2026). Applied ergonomics. *Purdue Road School,* West Lafayette, Indiana, US.

[P01] Guo, X., **Hu, X.,** Chen, Y., & Zhang, J. (2023, 2024, 2025, & 2026). Applied ergonomics. *Joint Transportation Research Program Poster Session,* Indianapolis, Indiana, US.

GRANTS

Total awarded grants: **$916,250**

GRANTS AWARDED

[G04] Title: SPR-5000: Retention of the Early-Career Highway Technician Workforce: Exploring and Addressing Their Career Trajectories and Challenges (10/01/2025 - 05/30/2027) – Indiana Department of Transportation **(INDOT)**

- Role: Research Assistant
- Responsibilities: (1) developed the research need statement; (2) prepared the full research proposal, including the outline, literature review, research design, work schedule, and formatting; and (3) advising a junior Ph.D. student (research assistant).
- Investigators: Dr. Yunfeng Chen (PI), Dr. Jiansong Zhang (Co-PI), & Dr. Steven Dunlop (Co-PI)
- Amount: $276,248

[G03] Title: SPR-4838: Automated Personalized Ergonomic Risk Report Generation System (08/01/2023 – 06/30/2028) – **INDOT**

- Role: Lead Research Assistant
- Responsibilities: (1) prepare meeting slides; (2) coordinate with INDOT district managers, subdistrict managers, and crew leaders; (3) conduct field data collection (physiological, motion, subjective, and environmental); (4) analyze data; and (5) prepare progress reports/papers.

- Investigators: Dr. Yunfeng Chen (PI), Dr. Yiheng Feng (Co-PI), & Dr. Jiansong Zhang (Co-PI)
- Amount: $283,508

[G02] Title: SPR-4806: INDOT Training Gap Analysis, Developing a Training Program for INDOT Workforce Development (new, Front-line and Supervisors) and Smart Tool for Effective Planning for Workforce (10/15/2023 - 10/14/2025) **– INDOT**

- Role: Research Assistant
- Responsibilities: (1) analyzed exit survey data collected by INDOT from departing employees using descriptive statistics, ANOVA, and post hoc analysis; and (2) prepared reports/papers/posters.
- Investigators: Dr. Steven Dunlop (PI), Dr. Satish Ukkusuri (Co-PI), & Dr. Yunfeng Chen (Co-PI)
- Amount: $256,665

[G01] Title: SPR-4523: Applied Ergonomics (08/24/2020 - 03/15/2023) **– INDOT**

- Role: Research Assistant
- Responsibilities: (1) analyzed electromyography, heart rate, and electrodermal activity data, together with subjective measures including rating of perceived exertion, usability, and perceived local pressure, using ANOVA and post hoc analysis; and (2) prepared papers/posters.
- Investigators: Dr. Yunfeng Chen (PI) & Dr. Jiansong Zhang (Co-PI)
- Amount: $99,829

UNFUNDED PROPOSALS

1. Title: Enhance Digital As-Builts with Materials Quality Assurance Data and Develop Pavement and Materials BIM Data Schema – Federal Highway Administration **(FHWA)**
   - Role: Research Assistant
   - Investigator: Dr. Yunfeng Chen (PI), Dr. Hubo Cai (Co-PI), Dr. Mohammad Khan (Co-PI), & Dr. Jiansong Zhang (Co-PI)
   - Responsibilities: prepared the full proposal, including the outline, literature review, research design, work schedule, and formatting.
   - Amount: $633,853.32
2. Title: A Guide for Holistic Information and Knowledge Management – National Cooperative Highway Research Program (**NCHRP)**
   - Role: Research Assistant
   - Investigator: Dr. Hubo Cai (PI), Dr. Yunfeng Chen (Co-PI), Dr. Jiansong Zhang (Co-PI), Dr. Lu Zhang (Co-PI), Scott Lecher (Consultant), Tanisha Hall (Consultant), & Kristen Jud (Consultant)

- Responsibilities: prepared the full proposal, including the outline, literature review, research design, work schedule, and formatting.
- Amount: $424,977

3. Title: Retention and Advancement of Women with Different Job Classifications in the Engineering and Transportation Workforce – **INDOT**
   - Role: Research Assistant
   - Investigator: Dr. Yunfeng Chen (PI)
   - Responsibilities: prepared the research need statement.

TEACHING EXPERIENCES

TEACHING INTERESTS

Construction Safety | Construction Financial Management | Human Factors in Construction and Transportation | Building Information Modeling | Construction Methods | Scheduling | Estimating

TEACHING ASSISTANT

Purdue University

CM 301 Introduction to Construction Company Financial Management Fall 2024 – Fall 2026
Fall 2026 (120 undergraduates); Fall 2025 (91); Spring 2025 (88); Fall 2024 (60)

- Delivered a lecture
- Assisted the instructor in developing the syllabus, exam questions, and other course materials
- Supported American Council for Construction Education (ACCE) accreditation by assessing student performance on Student Learning Outcomes (SLOs)
- Revised course materials to meet Americans with Disabilities Act (ADA) requirements
- Coordinated testing accommodations for students registered with the Disability Resource Center (DRC)
- Analyzed exam results and prepared performance summaries for the instructor
- Held office hours and answered student questions via email
- Graded homework with written feedback

CM 535 Construction Accounting and Financial Management Spring 2024, Spring 2026
Spring 2026 (42 graduate students); Spring 2024 (46)

- Mentored a junior Ph.D. student on grading and communicating with students
- Revised course materials to meet ADA requirements
- Answered student questions via email and conducted one-on-one post-exam reviews
- Graded homework with written feedback

CM 200 Intermediate Pre-Construction Management Fall 2022 – Spring 2024

Spring 2024 (70 undergraduates); Fall 2023 (87); Spring 2023 (98); Fall 2022 (98)

- Delivered lectures on project delivery methods
- Taught lab sections on construction company accounting
- Graded homework with written feedback

CM 581 Construction Finance and Company Management Spring 2023

Spring 2023 (11 graduate students)

- Answered student questions via email
- Analyzed exam results and prepared performance summaries for the instructor
- Graded homework

MENTORING

- Ze Wang, Ph.D. candidate, *Purdue University* Fall 2023 – Present
  − Role: Teaching mentor
  − Responsibilities: mentored teaching on how to grade assignments and communicate with students
- Bingzhen Li, Ph.D. student, *Purdue University* Fall 2025 – Present
  − Role: Research mentor
  − Responsibilities: mentored research on (1) retention of the Early-Career Highway Technician Workforce in INDOT; and (2) field assessment of occupational heat strain in highway maintenance workers via wearable physiological monitoring, including proposal writing, literature review, research design, data collection, data analysis, progress reports, and poster presentations
- Hongyu Tu and Yipeng Wei, Ph.D. students, *Purdue University* Fall 2026
  − Role: Research mentor
  − Responsibilities: mentored research on computer vision-based ergonomic risk evaluation, including data collection and data analysis
- Vivek reddy Puttha, master student, *Purdue University* Summer 2026
  − Role: Research mentor
  − Responsibilities: mentored on field data collection and data analysis for the ergonomics project [G03]

ACADEMIC SOCIETY SERVICES

PROFESSIONAL SOCIETIES

Member - *American Society of Civil Engineers* Fall 2023 – Present
Member - *American Society for Engineering Education* Fall 2026 – Present

JOURNAL REVIEWERS

- Human Factors and Ergonomics in Manufacturing
- Journal of Construction Engineering and Management

CONFERENCE REVIEWERS

- Associated Schools of Construction Annual Conference 2025, 2026
- Construction Research Congress 2026
- Transportation Research Board (TRB) Annual Meeting 2027

HONORS & AWARDS

Dean's Graduate Student Travel Award, *Purdue University*	2023 & 2024
Bowen School of Construction Graduate Student Travel Grant, *Purdue University*	2023 & 2024
Best Paper Award, 14th International Conference on Applied Human Factors and Ergonomics, 2023	2023
Postgraduate Academic Scholarship of Wuhan University, *Wuhan University*	2020 & 2021
Merit Student of Wuhan University, *Wuhan University*	2020
Excellent Teaching Assistant of Wuhan University, *Wuhan University*	2020
Excellent Undergraduate Graduation Thesis, *Wuhan University*	2019
Merit Student Model of Wuhan University, *Wuhan University*	2018
First Place, Wuhan University Student Structural Design Competition	2018
Second Place, Wuhan University Student Structural Design Competition	2016 & 2017
Third Prize, National MathorCup Mathematical Modeling Challenge	2018
Merit Student of Wuhan University, *Wuhan University*	2016 & 2017
Wuhan University Scholarship, *Wuhan University*	2016 - 2018

SKILLS

- Programming skills: Python, MATLAB
- Building Information Modeling: Revit
- Computer-Aided Design: AutoCAD, SolidWorks
- Hardware: wearable sensors (EMG, EDA, IMU such as Xsens), environmental sensors (vibration, air quality, noise, temperature, humidity)
- Statistical analysis: R, SPSS, SAS
- Modeling and animation: 3ds Max, Unity
- Musculoskeletal modeling: OpenSim
- Network analysis: Gephi, Netminer